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Characterizing multi-pollutant air pollution in China: Comparison of three air quality indices

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ABSTRACT

Multi-pollutant air pollution (i.e., several pollutants reaching very high concentrations simultaneously) frequently occurs in many regions across China. Air quality index (AQI) is used worldwide to inform the public about levels of air pollution and associated health risks. The current AQI approach used in China is based on the maximum value of individual pollutants, and does not consider the combined health effects of exposure to multiple pollutants. In this study, two novel alternative indices – aggregate air quality index (AAQI) and health-risk based air quality index (HAQI) – were calculated based on data collected in six megacities of China (Beijing, Shanghai, Guangzhou, Shjiazhuang, Xi'an, and Wuhan) during 2013 to 2014. Both AAQI and HAQI take into account the combined health effects of various pollutants, and the HAQI considers the exposure (or concentration)-response relationships of pollutants. AAQI and HAQI were compared to AQI to examine the effectiveness of the current AQI in characterizing multi-pollutant air pollution in China. The AAQI and HAQI values are higher than the AQI on days when two or more pollutants simultaneously exceed the Chinese Ambient Air Quality Standards (CAAQS) 24-hour Grade II standards. The results of the comparison of the classification of risk categories based on the three indices indicate that the current AQI approach underestimates the severity of health risk associated with exposure to multi-pollutant air pollution. For the AQI-based risk category of 'unhealthy', 96% and 80% of the days would be 'very unhealthy' or 'hazardous' if based on AAQI and HAQI, respectively; and for the AQI-based risk category of 'very unhealthy', 67% and 75% of the days would be 'hazardous' if based on AAQI and HAQI, respectively. The results suggest that the general public, especially sensitive population groups such as children and the elderly, should take more stringent actions than those currently suggested based on the AQI approach during high air pollution events. Sensitivity studies were conducted to examine the assumptions used in the AAQI and HAQI approaches. Results show that AAQI is sensitive to the choice of pollutant irrelevant constant. HAQI is sensitive to the choice of both threshold values and pollutants included in total risk calculation.

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1. Introduction

China has been experiencing serious air pollution problems in recent decades due to the rapid industrialization and urbanization, and increasing energy consumption. Numerous studies have demonstrated associations between air pollution and various health effects (e.g., Brunekreef and Forsberg, 2005; Burnett et al., 2000; Dockery, 2001; Dominici et al., 2005; Le Tertre et al., 2002). Recognizing the severity of the air pollution situation and the dense population in China, Chinese scientists have recently started air pollution-health effects studies (e.g., Cao et al., 2012; Chen, 2007; Chen et al., 2013; Guo et al., 2013; Kan and Chen, 2002; Kan and Gu, 2011). The results of these studies confirmed that air pollution threatens public health in China. Therefore, it is important to inform the public about the levels of air pollution and associated health risks so that people can take measures to protect their health.

Air pollution levels are determined by the concentrations of a complex mixture of air pollutants. Currently SO_2, NO_2, CO, O_3, $PM_{2.5}$ and PM_{10} are defined as the six criteria pollutants around the world in quantifying air pollution levels. The concentrations among the pollutants can be different by orders of magnitude, and their unit-concentration health effects are significantly different as well. Therefore, it is difficult for the general public to use the concentrations directly to characterize the levels of air pollution. Alternatively, the use of an index ranging from good to unhealthy is more understandable

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for the general public, and becomes the most common way to interpret air pollution levels in many countries and regions (Shooter and Brimblecombe, 2005). Such indices were firstly developed in early 1970's (Babcock, 1970), and have been evolving since then. Currently, Air Quality Index (AQI) is the most used index worldwide.

The most commonly used index is the United States Environmental Protection Agency (US EPA) AQI. The AQI ranges from 0 to 500 and is calculated based on the concentrations of the six criteria pollutants. For a given location on a given day, a sub-AQI for every pollutant is calculated, and the maximum of all the sub-AQIs is defined as the overall AQI. It indicates "unhealthy" air quality when AQI is greater than 100. This method has been criticized because it does not appropriately represent the combined health effects of exposure to multiple pollutants. A few studies have been conducted to develop alternative approaches that take into account the combined health effects of various pollutants. Swamee and Tyagi (1999) proposed to combine the sub-AQIs to form an Aggregate AQI (AAQI). Kyrkilis et al. (2007) followed this idea and developed an AAQI for the area of Athens, Greece. The results indicate that the AAQI estimates the air pollution exposure more effectively than the US EPA AQI. Another novel alternative AQI approach is based on the health risk associated with exposure to multiple air pollutants (Cairncross et al., 2007; Sicard et al., 2012; Stieb et al., 2008; Wong et al., 2013). The health-risk based AQI (HAQI) considers the established exposure (or concentration)-response relationships. HAQI has shown improvement over the existing AQIs in various countries and regions. Currently, Canada is using the Air Quality Health Index approach that is based on the approach proposed by Stieb et al. (2008).

The Chinese Ministry of Environmental Protection (MEP) adopted the US EPA AQI approach and developed the Chinese AQI system in 2012 (MEP, 2012a). Wong et al. (2013) developed a HAQI for Hong Kong, but no studies have been conducted to investigate the differences between the AQI, AAQI, or HAQI approaches in designating air pollution severities in mainland China. In a recent study, the spatial and temporal distributions of six criteria pollutants in China were revealed (Wang et al., 2014), making it possible for the first time to investigate how effective different AQI approaches in characterizing the severity of multiple-pollutant air pollution in China. In this study, AAQI was defined based on the sub-AQI of individual pollutant and HAQI was developed using the exposure-response relationships from the health studies in China. Both HAQI and AAQI were compared with the MEP's AQI in different seasons and locations. Health implications of the three indices under different pollution situation were also investigated. Hypotheses in each approach were discussed and future research needs to establish an optimal index that can provide the general public with the accurate information for the purpose of health protection were highlighted.

2. Methods

2.1. Study areas and data source

Six representative cities (Beijing, Shanghai, Guangzhou, Shijiazhuang, Xi'an, and Wuhan) were selected for comparative analyses of AQI, AAQI, and HAQI in China. The locations of the six cities are shown in Fig. 1. Beijing, Shanghai, and Guangzhou are located in the North China Plain, the Yangtze River Delta, and the Pearl River Delta, respectively. These regions have become hot spots of air pollution studies in China due to dense population, more developed economy, and frequent pollution events. A recent study investigating the air pollution status in China during 2013–2014 revealed that Shijiazhuang, Wuhan, and Xi'an are the most heavily polluted provincial capital cities in the North, South-East, and West China, respectively (Wang et al., 2014). The six cities located

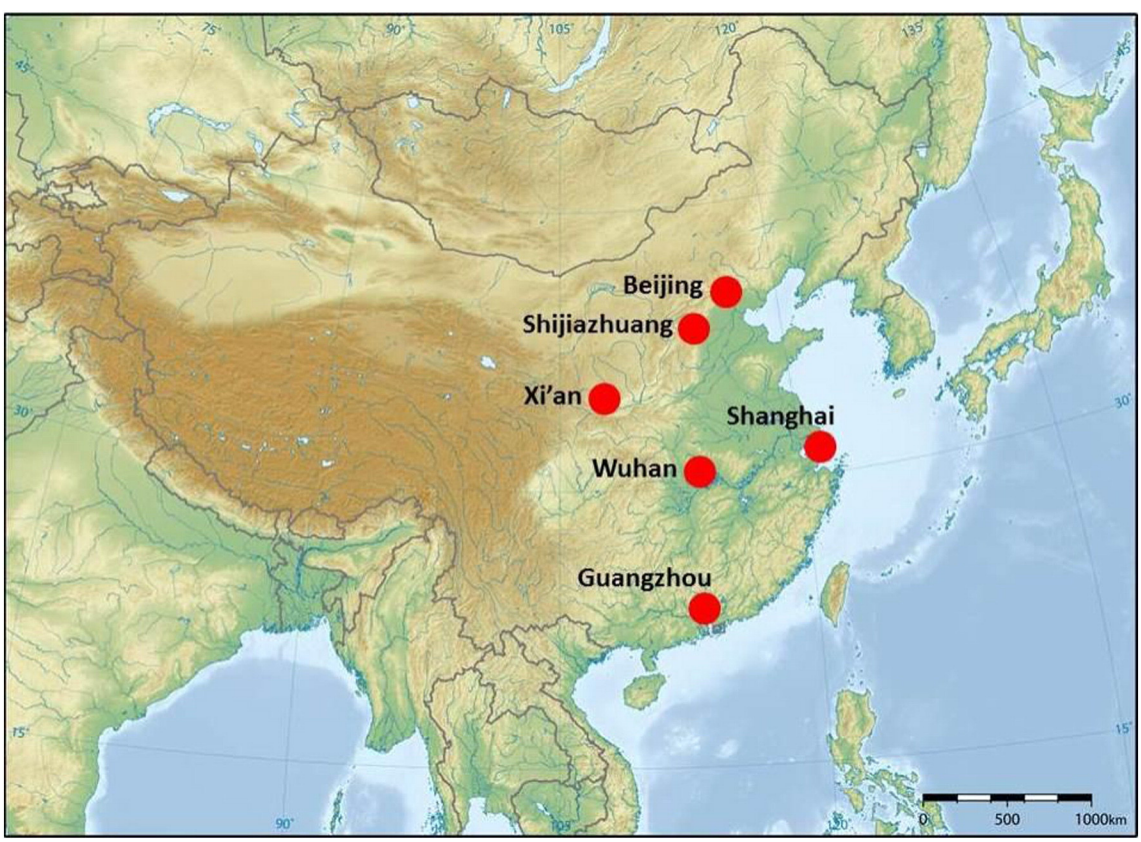

Fig. 1. Location of the six cities focused in the study. Beijing, Shanghai, and Guangzhou are the biggest and most developed cities located in the North China Plain, the Yangtze River Delta, and the Pearl River Delta, respectively. Shijiazhuang, Wuhan, and Xi'an are the most heavily polluted provincial capital cities in the North, South-East, and West China, respectively.

Table 1
Population, pollutants concentrations (mean ± 1 Standard Deviation), and non-attainment days of studied cities in China.

	Population[&]	Death rate	$PM_{2.5}$	PM_{10}	SO_2	NO_2	O_3[*]	CO	Pollution days[#]
	Million	‰	μg/m^3	μg/m^3	μg/m^3	μg/m^3	μg/m^3	mg/m^3	Days
CAAQS standards[$]			35/75	50/150	50/150	80/80	100/160	4/4	
City									
Beijing (BJ)	12.97	4.31	87 ± 67	109 ± 62	26 ± 23	51 ± 23	96 ± 58	1.3 ± 0.8	105/182/345
Shanghai (SH)	14.27	5.24	56 ± 41	80 ± 47	20 ± 14	41 ± 18	103 ± 45	0.9 ± 0.3	49/100/342
Guangzhou (GZ)	8.22	6.17	52 ± 28	72 ± 35	20 ± 9	49 ± 21	96 ± 51	1 ± 0.2	43/89/329
Shijiazhuang (SZ)	10.05	6.73	144 ± 109	287 ± 139	89 ± 60	64 ± 27	99 ± 62	1.8 ± 1.3	224/261/303
Xi'an (XA)	7.96	4.69	97 ± 83	181 ± 127	40 ± 29	51 ± 18	79 ± 43	2 ± 0.9	127/185/315
Wuhan (WH)	8.22	5.54	92 ± 70	135 ± 75	43 ± 20	53 ± 25	114 ± 54	1.3 ± 0.5	117/193/306

[&] Population is the permanent resident population; the temporary resident population is not included.
[*] O_3 standards are daily maximum 8-hour standards.
[$] Standards are 24-hour Grade I/II standards. The Grade II standards correspond to an AQI value of 100.
[#] The first number represents pollution days with at least two pollutant exceeding Grade II standards, the secondary number represents pollutions days with at least one pollutant exceeding the Grade II standards, and the third number represents total days with valid air quality monitoring data from 03/01/2013 to 02/28/2014.

in different regions exhibit significant differences in pollution types and levels (Table 1). Therefore, analyses of different AQI approaches using data from the six cities could provide an overall understanding on the effectiveness of different AQIs to characterize the air pollution exposures in China.

Hourly concentrations of SO_2, NO_2, $PM_{2.5}$ PM_{10}, CO, and O_3 in the six cities of interest were downloaded from the publishing website of China National Environmental Monitoring Center (http://113.108.142.147:20035/emcpublish/). One-year data from March 1st, 2013 to February 28th, 2014 are included in this study. This dataset has been used in previous studies (Hu et al., 2014; Wang et al., 2014; Zhang et al., 2015) to investigate the temporal and spatial variations in the gaseous and particulate pollutants in China. Detailed monitoring techniques have been provided in these studies, therefore, only a brief description of the dataset is provided here. All the measurements were conducted at the national air quality monitoring sites located in each city. At each site, automated monitoring systems have been installed and used to measure the ambient concentration of SO_2, NO_2, O_3, CO, $PM_{2.5}$, and PM_{10} according to China Environmental Protection Standards HJ 193–2013 (http://www.es.org.cn/download/2013/7-12/2627-1.pdf), and HJ 655–2013 (http://www.es.org.cn/download/2013/7-12/2626-1.pdf). The values from the monitoring sites at each city were automatically reported to the China National Environmental Monitoring Center and published after being validated based on Technical Guideline on Environmental Monitoring Quality Management HJ 630–2011 (http://kjs.mep.gov.cn/hjbhbz/bzwb/other/qt/201109/W020120130585014685198.pdf). The citywide average concentrations were calculated by averaging the concentrations at all sites in each city. This is the same method that the Chinese MEP uses to report daily concentrations of air pollutants to the public.

2.2. Air quality indices

2.2.1. MEP's AQI (or AQI)

The MEP's AQI approach is based on the Chinese Ambient Air Quality Standards (CAAQS). The sub-AQI of each pollutant is calculated using Eq. (1) with the monitored pollutant concentrations:

$$\begin{aligned} AQI_i &= \frac{(AQI_{i,j} - AQI_{i,j-1})}{(C_{i,j} - C_{i,j-1})} \times (C_{i,m} - C_{i,j-1}) + AQI_{i,j-1}, j>1 \\ AQI_i &= AQI_{i,1}\frac{C_{i,m}}{C_{i,1}}, j = 1 \end{aligned} \quad (1)$$

where AQI_i is the index of pollutant i; $C_{i,m}$ is the monitored ambient concentration of pollutant i; j is the health category index so that $C_{i,j}$ and $C_{i,j-1}$, which are the upper limit concentrations for the jth and j-1th health categories, encompass $C_{i,m}$; $AQI_{i,j}$ and $AQI_{i,j-1}$ are the AQI values of pollutant i corresponding to $C_{i,j}$ and $C_{i,j-1}$, respectively. Table 2 gives the health categories and the corresponding ranges of AQI values and pollutant concentrations. The overall AQI is the maximum of the sub-AQI of all pollutants, as shown in Eq. (2):

$$AQI = \max(AQI_1, AQI_2, \ldots, AQI_n), n = 1, 2, \cdots, 6. \quad (2)$$

The AQI approach converts the pollutant concentrations to a number on a scale of 0 to 500, with a sub-AQI of 100 corresponding to the upper limit of the CAAQS 24-hour Grade II standards (MEP, 2012b). The

Table 2
The health categories and meanings for AQI, the upper limit AQI values and corresponding pollutants concentrations based on the Technical Regulation on Ambient Air Quality Index (on trail) (http://kjs.mep.gov.cn/hjbhbz/bzwb/dqhjbh/jcgfffbz/201203/t20120302_224166.htm).

	Pollutant concentrations (μg/m^3)							
AQI below	SO_2	NO_2	CO	O_3	$PM_{2.5}$	PM_{10}	Health risk category	Meaning
50	50	40	2	100	35	50	Good	Satisfactory AQ, no risk
100	150	80	4	160	75	150	Moderate	Acceptable AQ, may be a moderate risk for a very small number of people who are unusually sensitive to certain pollutants
150	475	180	14	215	115	250	Unhealthy for sensitive groups	Children, older adults, and people with lung and heart disease are at a greater risk
200	800	280	24	265	150	350	Unhealthy	Everyone begins to experience some adverse health effects, and sensitive groups experience more serious effects
300	1600	565	36	800	250	420	Very unhealthy	Everyone experience more serious health effects
400	2100	750	48	1000	350	500	Hazardous	Entire population is affected, stay indoors and avoid outdoor activities.
500	2620	940	60	1200	500	600	Severe	Entire population is affected, stay indoors and avoid outdoor activities.

air quality level at a location on a given day is defined as unhealthy if the overall AQI level is greater than 100 based on daily-average concentrations. This AQI calculation approach is referred as AQI in Results and discussion.

2.2.2. Aggregate AQI (AAQI)

The AAQI approach takes into account the combined effects of all criteria pollutants. The formula proposed by Swamee and Tyagi (1999) and Kyrkilis et al. (2007) was adopted to define the AAQI as shown in Eq. (3):

$$AAQI = \left(\sum_{i=1}^{n}(AQI_i)^{\rho}\right)^{\frac{1}{\rho}} \quad (3)$$

where ρ is a pollutant irrelevant constant. In an extreme case of $\rho = \infty$, the AAQI equals the maximum of sub-AQI indices of all criteria pollutants, which becomes the AQI approach described in the previous section. In the other extreme case of $\rho = 1$, the AAQI becomes the linear summation of all sub-AQI indices. The selection of the most proper value for ρ is still an open scientific issue (Kyrkilis et al., 2007). Previous studies suggested values of ρ should be between 2 and 3 (Khanna, 2000; Swamee and Tyagi, 1999). In this study, the value $\rho = 2$ was adopted, and the effect of choosing different ρ values on AAQI is discussed in Results and Discussion. The same scale of 0–500 is applied to for better comparison with AQI (and HAQI described in the following section). Therefore, AAQI (and HAQI) values are capped at 500 when they exceed 500.

2.2.3. Health-risk based AQI (HAQI)

While the AAQI attempts to account for the combined health effects of multiple air pollutants, it does not explicitly account for the established exposure-response relationships of the pollutants. Health-risk based indices were proposed in a few studies to include such exposure-response characteristics (Cairncross et al., 2007; Sicard et al., 2012; Stieb et al., 2008; Wong et al., 2013). To define a health-risk based AQI, the total excess risk (ER) of exposure to multiple pollutants based on the Cairncross's concept (Cairncross et al., 2007) can be used. In a general form, relative risk (RR) for each pollutant is estimated based on health effect studies, using Eq. (4):

$$RR = \exp[\beta(C - C_0)],\ C > C_0 \quad (4)$$

where β is the exposure-response relationship coefficient, representing the excess risk of health effect (such as mortality) per unit increase of pollutant (such as 1 μg/m^3 of $PM_{2.5}$); C is the concentration of pollutant, and C_0 is the threshold concentration, below which it is believed the pollutant demonstrates no obvious adverse health effects (i.e. RR = 1). For this study, β values were chosen from a systematic review of Chinese studies on short-term exposure to air pollution and daily mortality of all ages (Shang et al., 2013). The β values are 0.38%, 0.32%, 0.81%, 1.30%, and 0.48% per 10 μg/m^3 increase of $PM_{2.5}$, PM_{10}, SO_2, NO_2 and O_3, respectively, and the β value is 3.7% per 1 mg/m^3 increase of CO.

It is presumed that the air pollution poses little or no health risk when concentrations of pollutants are below the Grade II standards. Therefore, the upper limits of CAAQS 24-hour Grade II standard for the six pollutants were used as the C_0 values as a first estimation. However, some studies indicated these values cannot be regarded as threshold values below which a zero adverse response may be expected (WHO, 2005), thus two sensitivity analyses were conducted by changing C_0 values to 1) the upper limits of CAAQS 24-hour Grade I standards and 2) no threshold values (i.e., $C_0 = 0$) except CO. The results of the sensitivity analyses are included in Results and discussion.

ER of each pollutant, which is defined as RR-1, is summed up to calculate the total excess risk (ER_{total}) for simultaneous exposure to several air pollutants:

$$ER_{total} = \sum_{i=1}^{n} ER_i = \sum_{i=1}^{n}(RR_i - 1). \quad (5)$$

Therefore, higher values of ER_{total} reflect higher health risks. Fig. 2 shows the average ERs of the six pollutants in the six cities using the chosen β values, the measured daily concentrations, and the C_0 values of the CAAQS 24 h Grade II standards. ER_{total} varies significantly among the six cities, with the minimum of 0.6% in Guangzhou and the maximum of 9.9% in Shijiazhuang. $PM_{2.5}$ contributes to ER_{total} the most in Beijing, Shanghai, and Wuhan; PM_{10} contributes to ER_{total} the most in Shijiazhuang and Xi'an, followed by $PM_{2.5}$; O_3 and NO_2 also have certain contributions, especially in Guangzhou. O_3, NO_2 and $PM_{2.5}$ have similar ER values. Note linearly adding up the pollutants' risks could over-estimate ER_{total} if certain pollutants are highly correlated, such as $PM_{2.5}$ and PM_{10}. Therefore, adding the six pollutants' ER is an upper bound estimation of ER_{total}. Analysis was conducted to estimate the ER_{total} by adding ERs of $PM_{2.5}$, O_3, and NO_2 in Results and Discussion.

Previous studies converted ER_{total} to an arbitrary index, ranging from 0–10, for warning the public about the air pollution risks (Cairncross et al., 2007). Such an index cannot be directly compared to AQI and AAQI as they are on different scales. In this study, the HAQI, with the same scale as AQI and AAQI (0–500), is introduced based on ER_{total} and the equivalent pollutant concentrations. The equivalent concentration (C^*) of a criteria pollutant is defined as the concentration level at which the ER of the pollutant equals to ER_{total}, as shown in Eq. (6):

$$RR^* = ER_{total} + 1 = \exp[\beta(C^* - C_0)] \quad (6)$$

where RR^* represents the relative risk calculated based on the equivalent pollutant concentration. Using the RR^* value determined from ER_{total} using Eq. (5), the equivalent pollutant concentration of the ith criteria pollutant (C_i^*) can be determined using Eq. (7):

$$C_i^* = \ln(RR^*)/\beta_i + C_{0,i} \quad (7)$$

where β_i and $C_{0,i}$ are the β and C_0 values of the ith pollutant, respectively. Because RR^* is the total risk of exposing to multi-pollutants, the

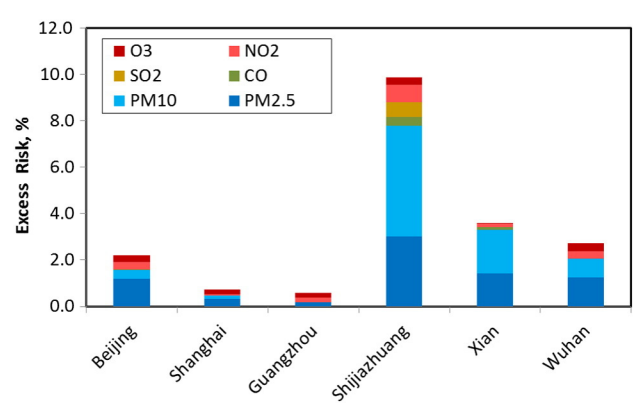

Fig. 2. The average excess risks (ER) of the six criteria pollutants in the six cities. Daily ER values are calculated based on relative risk (RR) using Eq. (4) with daily concentrations in each city, and then is averaged over the 1-year period. In the calculation, β values were chosen from a systematic review of Chinese studies of short-term exposure to air pollution and daily mortality of all ages (Shang et al., 2013) and C_0 value of each pollutant was the upper limit of its CAAQS 24-hour Grade II standard.

equivalent concentration represents the combined health risk of all criteria pollutants and is naturally higher than the actual concentration of the pollutant. The HAQI based on the equivalent concentration of the ith criteria pollutant (sub-HAQI, or $HAQI_i$) can then be determined using C_i^* as shown in Eq. (8), instead of the actual concentration C_i in Eq. (1),

$$HAQI_i = \frac{(AQI_{i,j} - AQI_{i,j-1})}{(C_{i,j} - C_{i,j-1})} \times (C_i^* - C_{i,j-1}) + AQI_{i,j-1}, j>1$$
$$HAQI_i = AQI_{i,1}\frac{C_i^*}{C_{i,1}}, j = 1. \quad (8)$$

The overall HAQI is determined as the maximum of all sub-HAQIs, as shown in Eq. (9)

$$HAQI = \max(HAQI_1, HAQI_2, \ldots, HAQI_n), n = 1, 2, \cdots, 6. \quad (9)$$

HAQI includes the information of exposure to multi-pollutants and considers the exposure-response relationships of different pollutants. HAQI is a function of C_0 and ER_{total}. Its value changes if a different set of C_0 values is chosen or different pollutants are used to calculate the ER_{total}. Impact of different choices of these parameters on HAQI is further discussed in Results and discussion.

Note that although Eq. (9) appears to be similar to Eq. (2), they are very different from the health risk perspective. Eq. (2) implies that the overall health risk of exposure to criteria pollutants is determined only by the pollutant that has the highest AQI value. However, each sub-HAQI in Eq. (9) already considers the relative risk due to multiple pollutants. Choosing the maximum sub-HAQI as the overall HAQI is a more cautious way of estimating the multi-pollutant health risk, considering the fact that the indices are going to be used to warn the general public, especially the sensitive population, of potential air pollution exposure.

It should also be noted that the AAQI and HAQI methods assume that effects of individual pollutants are additive. However, non-additive

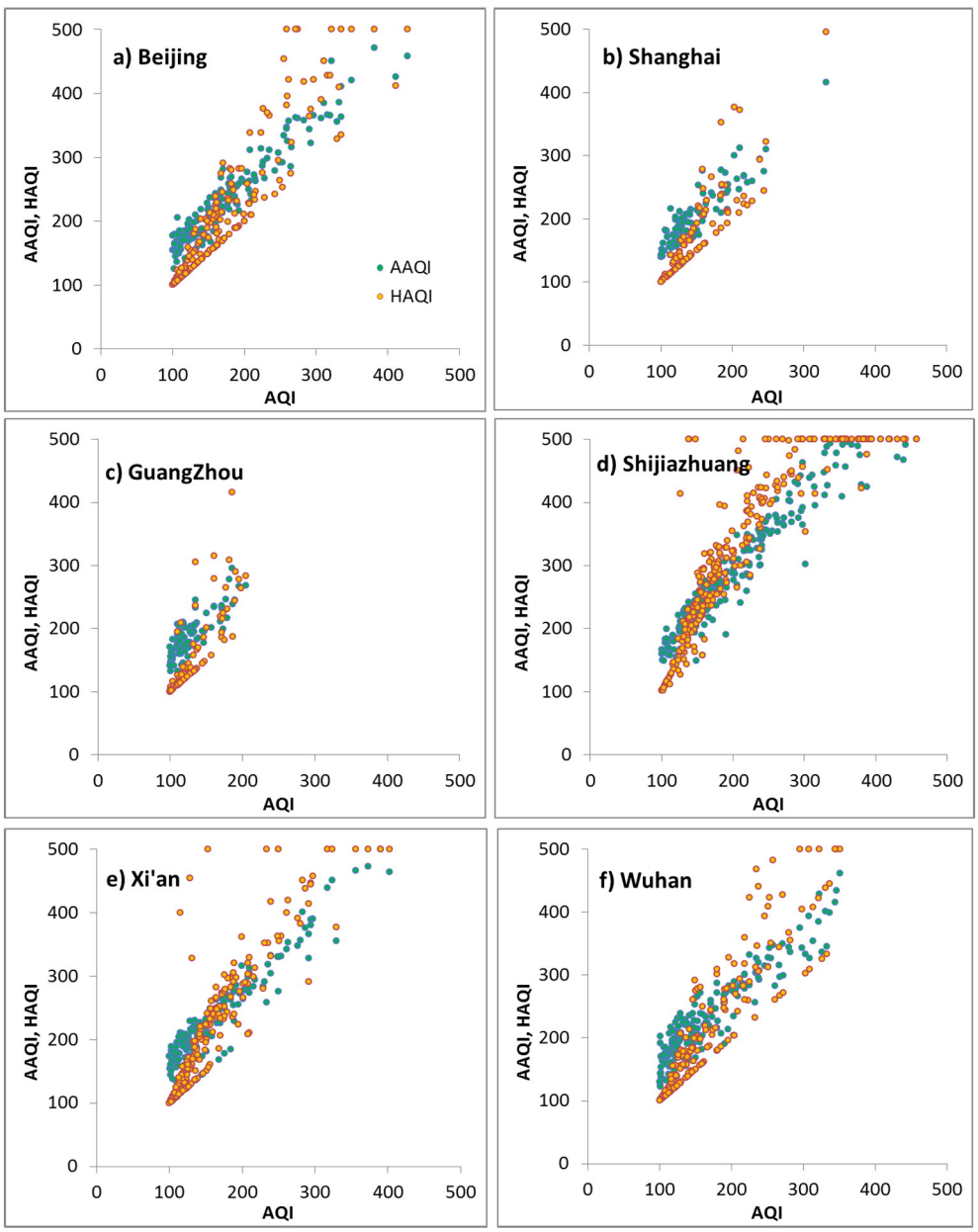

Fig. 3. Comparison of AAQI and HAQI to AQI in the six cities on unhealthy days (i.e., AQI > 100). The AAQI and HAQI are designed to have the same scale as AQI, i.e. 0–500. Therefore, AAQI and HAQI are capped at 500 when their calculated values exceed 500.

interactions (such as synergism or antagonism) among pollutants might occur in reality (Mauderly et al., 2010). In the past years, various methods, such as the statistical interaction approach, the indicator approach, the source apportionment techniques and risk-based metrics, have been developed to estimate total health risk of multi-pollutant exposure (Dominici et al., 2010; Oakes et al., 2014). Despite the progress in characterizing multi-pollutant exposure metrics, the actual interaction among pollutants remains unclear and additional research is needed to quantify the combined effects of multiple pollutants.

3. Results and discussion

Fig. 3 shows the comparison of AAQI and HAQI to AQI on unhealthy days (i.e., when AQI > 100). In all cities, AAQI is always higher than AQI. HAQI equals to AQI on healthy days (i.e., when AQI ≤ 100) and on single-pollutant unhealthy days (i.e., when AQI > 100 and only one pollutant exceeds its CAAQS 24-hour Grade II standard). HAQI is higher than AQI on multi-pollutant unhealthy days (i.e., when AQI > 100 and more than one pollutants exceed the CAAQS 24-hour Grade II standards). Table 1 shows that of all the unhealthy days during 2013–2014, 58%, 49%, 48%, 86%, 69%, and 61% of the days are with at least two pollutants simultaneously exceeding the CAAQS 24-hour Grade II standards in Beijing, Shanghai, Guangzhou, Shijiazhuang, Xi'an and Wuhan, respectively. The fact that AQI is lower than both AAQI and HAQI on these days indicates that the current AQI approach likely under-reports the severity of the health risk associated with exposure to multi-pollutant air pollution.

On unhealthy days, AAQI is highly correlated with AQI in all cities, with correlation coefficients (R^2) in the range of 0.73 (Guangzhou) to 0.93 (Shijiazhuang). HAQI has weaker correlation with AQI, compared to AAQI. R^2 of HAQI to AQI is from 0.69 (Guangzhou) to 0.85 (Beijing). Fig. 3 shows that AAQI is generally greater than HAQI on the "unhealthy for sensitive groups" days (i.e., 100 < AQI ≤ 150), but HAQI generally becomes greater on the "very unhealthy" days (i.e., when AQI ≥ 200). The results suggest that the AAQI approach, without considering the actual pollutants' exposure-response relationships, could still underestimate the health risks on these high pollution days, even though it takes into account the combined effects of all pollutants.

Air quality is divided into six health risk categories based on the AQI value and certain actions are advised at the different health risk levels to protect health (Table 1). Misclassification of air pollution levels will result in inappropriate protection recommendations to the public and consequently cause potential health problems. Fig. 4 shows the total number of days (sum of the six cities) of five risk categories based on AQI (0–100 good-moderate, 101–150 unhealthy for sensitive groups, 151–200 unhealthy, 201–300 very unhealthy, and >300 hazardous). The classification of the category on a certain day could change if based on AAQI or HAQI. Figs. 4(a) and (b) show the number of days in each of the AQI-based category changes into different AAQI-based and HAQI-based categories, respectively. For the AQI-based risk category of 'unhealthy for sensitive groups', 95% of the days would be 'unhealthy' or even higher risk category if based on AAQI, and the ratio would be 33% if based on HAQI; for the AQI-based risk category of 'unhealthy', 96% and 80% of the days would be at least one risk category higher if based on AAQI and HAQI, respectively; and for the AQI-based risk category of 'very unhealthy', 67% and 75% of the days would be 'hazardous' if based on AAQI and HAQI, respectively. The results show that the actual health risk is likely greater than what the current AQI suggests because the public is exposed to multiple pollutants on most pollution days, while AQI reflects only the single pollutant with the 'greatest health risk', neglecting the health risks of other pollutants.

Fig. 5(a) shows the mean values and standard deviations of the ratios of AAQI and HAQI to AQI in the six cities on all days. Substantial differences are found between AAQI and HAQI. The mean HAQI/AQI ratios range from 1.04 (Shanghai) to 1.46 (Shijiazhuang). The mean AAQI/AQI ratios are generally larger than HAQI/AQI ratios, ranging

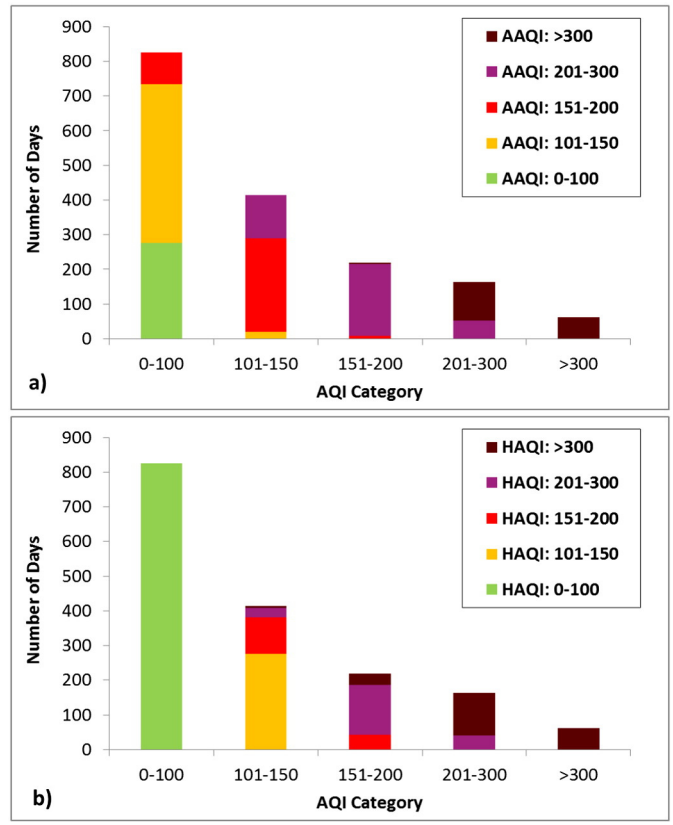

Fig. 4. Classification of air pollution health risk categories using the AQI approach, comparing to the classifications based on a) AAQI and b) HAQI. All data from the six studied cities were included in the analysis.

from 1.46 (Beijing) to 1.63 (Guangzhou). The difference in the mean AAQI/AQI ratios among the six cities is much less than in HAQI/AQI ratios. The standard deviations of AAQI/AQI ratios range from 0.16 (Shijiazhuang) to 0.24 (Wuhan), showing small variations among the cities. However, the standard deviations of HAQI/AQI ratios show much greater variations, ranging from 0.13 (Shanghai) to 0.38 (Shijiazhuang). The results reveal large discrepancy resulted from different approaches to characterize the multi-pollutant air pollution, and merit further investigation to identify a more accurate index.

The pollutant irrelevant constant ρ is the key parameter in AAQI calculation. Previous studies suggested values of ρ should be between 2 and 3 (Khanna, 2000; Swamee and Tyagi, 1999), but the most proper value for ρ is still under debate. A ρ value of 2.0 was firstly chosen in the AAQI calculation and sensitivity study was further conducted to examine the effect of ρ on the AAQI results by choosing ρ values of 1.5, 2.5, and 3.0. Fig. 5(b) shows the comparison of mean and standard deviation of AAQI/AQI ratios with the four different ρ values. The results indicate that the choice of the ρ values greatly influences the AAQI values. AAQI decreases with the increase of the ρ value. The AAQI/AQI ratios show small variations among the six cities, ranging from 1.85 to 2.1 with $\rho = 1.5$, 1.46 to 1.63 with $\rho = 2.0$, 1.30 to 1.42 with $\rho = 2.5$, and 1.21 to 1.30 with $\rho = 3.0$, despite the quite different pollution status in the six cities. This result indicates that AAQI is not sufficient to distinguish the risks due to different air pollution conditions.

HAQI calculation is dependent on C_0 and ER_{total}. The CAAQS 24-hour Grade II standards were used firstly as C_0. However, some studies indicated that guideline values cannot be regarded as threshold values below which a zero adverse response may be expected (Bell et al., 2006; Gent et al., 2003). Some studies indicated that PM_{10}, $PM_{2.5}$, O_3, SO_2 and NO_2 may not have apparent threshold values below which the risk of adverse health effect is zero (WHO, 2005). Therefore, two

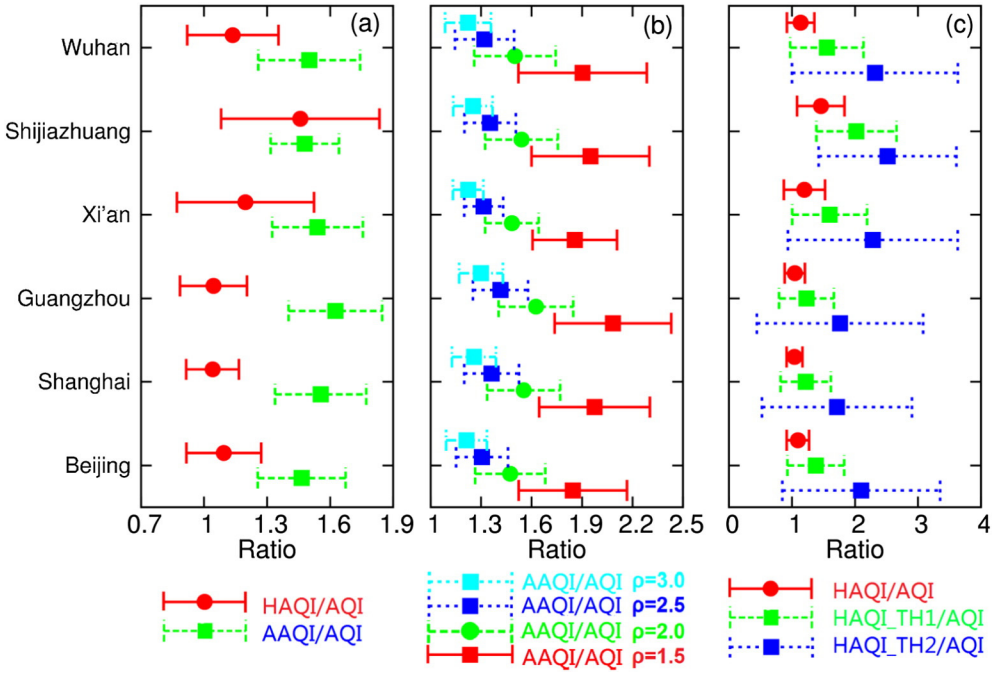

Fig. 5. (a) Mean and standard deviation of HAQI/AQI (red) and AAQI/AQI (green) ratios in the six cities; (b) mean and standard deviation of AAQI/AQI ratios when choosing ρ values of 1.5, 2.0, 2.5, and 3.0; and (c) mean and standard deviation of HAQI/AQI ratios when setting C_0 values to the CAAQS 24 h Grade II standards (HAQI/AQI, red), the CAAQS 24 h Grade I standards (HAQI_TH1/AQI, green), and zero except CO (HAQI_TH2/AQI, blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sensitivity studies were conducted to examine how the threshold values affect the HAQI values. In the first sensitivity study, the CAAQS 24-hour Grade I standards were used as the C_0 values to calculate HAQI (named as HAQI_TH1); and in the second one, zero C_0 values for all pollutants except CO (which C_0 value was kept at 4 mg/m^3) were used to calculate HAQI (named as HAQI_TH2). Fig. 5(c) compares the mean ratios and standard deviations of HAQI_TH1, HAQI_TH2, and the original HAQI to AQI. The HAQI is sensitive to the choice of threshold values. Lower C_0 values result in higher HAQI. The HAQI_TH1/AQI, HAQI_TH2/AQI ratios are 20–40% and 50–90% higher than the HAQI/AQI ratios, respectively. The large difference caused by the choice of C_0 values highlights the research need of the threshold values of all pollutants for informing the general public of air pollution health effect and protecting public health.

Another assumption of ER_{total} was also examined. ER_{total} was firstly calculated as the sum of the individual risk of all pollutants. However, some pollutants were found to be highly correlated, such as $PM_{2.5}$ and PM_{10} (Wang et al., 2014). Linear summation of all pollutants' risks directly over-estimates the total risk. The HAQI values can therefore be treated as the upper bound estimation by summing up risks of all pollutants. Some studies considered different pollutants to calculate the total risk. For example, the Canadian Air Quality Health Index is calculated based on the individual risk of $PM_{2.5}$, O_3, and NO_2 (https://ec.gc.ca/cas-aqhi/). Therefore, in this study, the HAQI based on the combination of $PM_{2.5}$, O_3 and NO_2 was also calculated and considered as an estimation of the lower bound of HAQI. Fig. 6 shows the monthly variations of HAQI (lower bound to upper bound), AAQI and AQI in the six cities. In all cities, the three indices have similar monthly variation, with the highest indices in the winter and the lowest in the summer. The HAQI range is larger in the winter than in the summer, because PM_{10}, SO_2, and CO concentrations are generally also high and contribute to the total risks in the winter. Shijiazhuang has the largest HAQI range among the six cities as it has the highest PM_{10} and SO_2 (Table 1). AAQI is generally higher than or close to the upper bound estimate of HAQI. The lower bound estimate of HAQI is close to AQI, mainly because O_3 levels are low when $PM_{2.5}$ levels are high especially in the winter, O_3 levels are high in the summer but $PM_{2.5}$ levels are low, and NO_2 levels are generally low compared to the CAAQS Grade II standard (Table 2). The lower bound of HAQI is lower than AQI when PM_{10} is the major pollutant while its risk is not included in the total risk calculation, which mostly occurs in Shijiazhuang and Xi'an. Wang et al. (2014) revealed significant fraction of coarse PM in China, especially in the western and northern China. Therefore, when estimating the total risk, PM_{10} or at least the coarse PM particles should be considered in the health risk estimation, in addition to the $PM_{2.5}$ risk.

The comparison indicates that during multi-pollutant air pollution events, when concentrations of several pollutants are simultaneously high and cause risk to public health, the current AQI approach underestimates the health risk. A combined index considering multi-pollutant exposure provides a more accurate description of the air pollution health risk in China than the current AQI. The findings in this study have important health implications. Results of AAQI and HAQI both indicate that the health risk is mostly 'at least one category' higher than what the current AQI suggests on days of AQI greater than 150. Based on this finding, the public, especially the sensitive groups (children, older adults, people with lung and heart diseases), should take more stringent actions to avoid the adverse effects of air pollution. This is especially true for heavily polluted cities and in the winter season when the pollution level is higher.

Even though the HAQI approach developed in this study is sensitive to the parameter of C_0 and approach to calculated ER_{total}, the main finding that health risk of exposure to multi-pollutant pollution is likely higher than what the current AQI suggests still holds true. The HAQI developed in this study is not aimed to replace the current AQI, but can serve as an alternative approach. HAQI considers the specific exposure-response relationships of multi-pollutants and therefore delivers more accurate pollution risk information. To identify an optimal HAQI approach that best captures the exposure-health associations in China, further investigation is needed on several aspects. As previously discussed, studies on the pollutants' threshold values and the total risk of multi-pollutant exposure are needed. Studies on associations between HAQI and observed health effects are also needed. Some

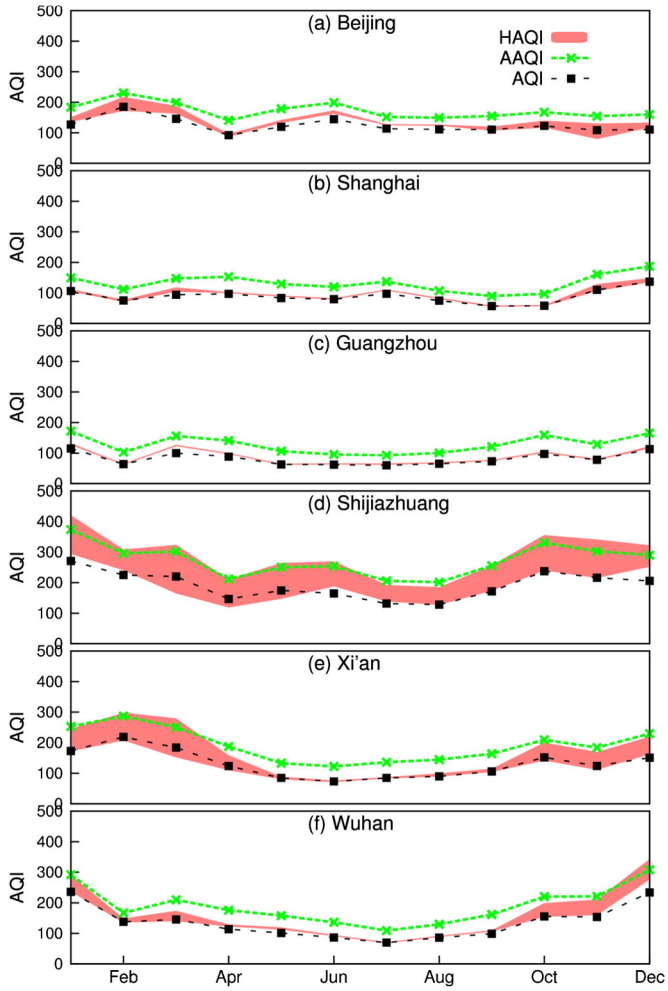

Fig. 6. Monthly variations of AQI, AAQI, and HAQI in the six cities. January and February are of 2014, and March to December are of 2013. The width of HAQI shadow is defined by two HAQI estimates (upper and lower bounds) choosing different combination of pollutants for the total excess risk calculation. HAQI using all six pollutants for the total excess risk calculation is treated as the upper bound, and using a subset of pollutants of $PM_{2.5}$, O_3, and NO_2 is treated as the lower bound.

studies found that HAQI (in different forms from the HAQI in this study) is significantly associated with observed health effects (Hong et al., 1999; To et al., 2013). Such studies should be conducted in all of China to examine the effectiveness of any proposed AQI approach in characterizing air pollution risks.

4. Conclusions

The AAQI and HAQI are higher than AQI on days when two or more pollutants simultaneously exceed the CAAQS 24-hour Grade II standards. The health risk categories are commonly used to inform the general public about the air pollution levels. The comparison result suggests that for the AQI-based risk category of 'unhealthy', 96% and 80% of the days would be 'very unhealthy' or 'hazardous' if based on the AAQI and HAQI, respectively; and for the AQI-based risk category of 'very unhealthy', 67% and 75% of the days would be 'hazardous' if based on the AAQI and HAQI, respectively. Therefore, the current AQI approach underestimates the severity of the health risk associated with the exposure to multi-pollutant air pollution. During high air pollution events, the health risk of air pollution is more serious relative to what the current AQI suggests, and the public, especially sensitive groups, should take more stringent actions to protect health. A significant difference is found between the three indices, and results from sensitivity studies show that the AAQI is affected by the choice of pollutant irrelevant constant ρ and the HAQI is sensitive to the choice of the threshold values and the pollutants included in the total risk calculation. Future studies should focus on the investigation of the pollutants' threshold values, the total risk of multi-pollutant exposure, and associations between the indices and observed health effects.

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